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from PIL import Image, ImageFilter
from google.colab.patches import cv2_imshow
import cv2

# Input image path
image_path = "lab_image.jpg"

# Open image
im1 = Image.open(image_path)

# Apply Unsharp Masking
im2 = im1.filter(
    ImageFilter.UnsharpMask(
        radius=3,
        percent=200,
        threshold=5
    )
)

# Save output image
im2.save("Unsharp_Masked_Image.jpg")

# Display images
original = cv2.imread("lab_image.jpg")
sharpened = cv2.imread("Unsharp_Masked_Image.jpg")

print("Original Image")
cv2_imshow(original)

print("Sharpened Image using Unsharp Masking")
cv2_imshow(sharpened)
```

Original Image



Sharpened Image using Unsharp Masking



